

FinTrust Digital Bank

Week 1: Business & Data Intelligence Assessment

Track: Data Analytics

Name: Adegbola Yusuf

Programme: AnalystLab Africa Experience Lab Internship

1. Introduction

For Week 1 of the FinTrust Digital Bank project, my focus was on understanding the business problem and becoming familiar with the data before moving into deeper analysis.

FinTrust is a fictional digital bank that wants to use its available data to better understand customer and transaction behaviour, support risk-related decisions, improve operations and strengthen its digital banking services.

As part of the Data Analytics track, my role is to determine how the available customer and transaction data can be turned into useful business intelligence. This week, I reviewed the supplied datasets, checked their structure and quality, identified useful analytical questions and KPIs, and planned the dashboard and analysis work that will continue during Weeks 2–4.

2. Business Understanding

FinTrust has customer and transaction data available, but raw records alone are not enough to support good business decisions. Management needs a

clearer view of what customers are doing, how transactions are performing, which banking channels are being used, and where operational or risk-related patterns may require attention.

The role of data analytics in this project is therefore to turn the available data into information that is easier to understand and act on.

For example, rather than looking through thousands of individual transaction records, management should be able to answer questions such as:

- Which customer groups generate the most transaction activity?
- Which transaction channels are used most often?
- What proportion of transactions are successful or failed?
- Which transaction types account for the most value?
- Where are risk-review flags most commonly found?

These types of insights can support areas such as transaction monitoring, channel management, customer engagement and risk-related investigation.

The analytical outputs may be useful to executive management, operations teams, digital banking or product teams, risk and compliance teams, and customer-support teams.

The Data Analytics track can contribute by profiling the available data, defining clear KPIs, analysing customer and transaction patterns and later developing a Power BI dashboard that presents the most important findings in a simple way.

The assignment defines the role of the Data Analytics track as determining how FinTrust's customer and transaction data can be transformed into useful business intelligence.

3. Resources Reviewed

I worked with three main data resources during Week 1:

Resource	What it contains	How I used it
FinTrust Customer Data	Customer demographics, segment, account information, engagement and preferred channel	Customer profiling and segmentation planning
FinTrust Transaction Data	Transaction amounts, types, channels, statuses, devices, locations and risk-review flags	Transaction profiling and KPI planning
FinTrust Data Dictionary	Definitions, data types and intended use of fields	Used to understand the correct meaning of each variable

The assignment identifies these three files as the approved data resources for the project.

The Customer dataset contains **1,500 customer records**, while the Transaction dataset contains **12,000 transaction records**.

The two datasets are linked through **Customer ID**

One important limitation is that all of the data is synthetic and was created for training purposes. The project notes also state that **Risk_Review_Flag** should be treated as a review indicator rather than confirmed fraud.

Another data-quality issue identified during profiling is that Device_Type and Location each contain **96 missing values**, which represents approximately **0.8%** of transaction records.

These missing records will be handled carefully during later analysis rather than being removed automatically.

4. Track Objectives

For the Data Analytics track, my main objectives are to:

- Understand the structure and quality of the customer and transaction datasets before carrying out deeper analysis.
- Develop useful analytical questions that can help FinTrust understand customer behaviour, transaction activity, channel usage, transaction outcomes and risk-related patterns.
- Define a clear set of KPIs that can be calculated consistently from the available data.
- Plan a simple and useful dashboard structure that can later be developed in Power BI.
- Create a practical analysis plan for Weeks 2–4 covering data preparation, exploratory analysis, dashboard development and final reporting

5. Success Criteria

I will consider the analytical work successful if:

- the datasets are correctly profiled and their structure, relationships and data-quality issues are clearly documented;
- the analytical questions can actually be answered using the available data;
- the proposed KPIs have clear definitions and can be calculated consistently;
- the dashboard plan connects directly to the business questions and does not include unnecessary visuals;

- the findings are based on evidence from the data and do not overstate what the data can prove;
- the Week 1 work provides a clear foundation for the analysis and dashboard work planned for Weeks 2–4.

6. Data Understanding and Profiling

6.1 Customer Dataset

The Customer dataset contains **1,500 records and 12 fields**.

The fields include:

Customer_ID, Customer_Name, Age, Gender, City, Customer_Segment, Account_Type, Tenure_Months, Digital_Engagement_Score, Monthly_Income_Band, Preferred_Channel, and Account_Status.

The main numerical fields are:

- Age
- Tenure_Months
- Digital_Engagement_Score

The main categorical fields are:

- Gender
- City

- Customer_Segment
- Account_Type
- Monthly_Income_Band
- Preferred_Channel
- Account_Status

My initial checks showed that the customer data is fairly clean:

- no missing values;
- no duplicate rows;
- all 1,500 Customer_ID values are unique;
- no obvious category-format inconsistencies were found;
- the engagement score values remain within the documented range.

This confirms that the customer dataset has a grain of **one row per customer**.

6.2 Transaction Dataset

The Transaction dataset contains **12,000 records and 11 fields**.

The fields include:

Transaction_ID, Customer_ID, Transaction_DateTime, Transaction_Type, Amount_NGN, Channel, Device_Type, Location, International_Transaction, Transaction_Status, and Risk_Review_Flag.

The main numerical field is:

- Amount_NGN

The main categorical fields include:

- Transaction_Type
- Channel
- Device_Type
- Location
- International_Transaction
- Transaction_Status
- Risk_Review_Flag

The main date/time field is:

- Transaction_DateTime

The transaction data covers the period from **January to March 2026**.

My profiling checks showed:

- no exact duplicate transaction rows;
- all 12,000 Transaction_ID values are unique;
- no zero or negative transaction amounts were identified;
- all transaction dates could be converted successfully;
- Device_Type has 96 missing values;
- Location has 96 missing values.

The missing values in Device_Type and Location each represent about **0.8%** of the transaction records.

6.3 Relationship Between the Datasets

The two datasets are connected using Customer_ID.

The data dictionary defines Customer_ID in the transaction table as a foreign key to the customer table.

The relationship is:

One customer can have many transactions.

I also checked the relationship directly and found that all **12,000 transaction records** successfully match a valid customer record.

This means the two datasets can be combined later to answer questions such as:

- transaction value by customer segment;
- success rate by account type;

- channel usage by customer characteristics;
- risk-review patterns by customer group.

6.4 Initial Data-Quality Observations

The main data-quality observations are:

Area	Observation
Customer missing values	None found
Customer duplicates	None found
Transaction duplicates	None found
Primary keys	Customer_ID and Transaction_ID are unique
Missing transaction data	96 missing Device_Type values and 96 missing Location values
Date/time quality	All transaction timestamps could be parsed
Amount validity	No zero or negative amounts found

Dataset relationship	All transactions match a valid customer
Risk field interpretation	Risk_Review_Flag is not confirmed fraud

7. Analytical Questions

After reviewing the datasets, I developed a set of questions that I believe will guide the analysis in the next stage of the project.

#	Analytical Question	Why It Matters
1	Which customer segments generate the highest transaction activity and value?	Helps FinTrust understand which customer groups contribute most to activity
2	How do transaction volume and transaction value change over time?	Helps identify trends, peaks or periods that may require further investigation
3	Which transaction channels are used most often and process the highest value?	Helps FinTrust understand how customers are using its banking channels
4	How do transaction success and failure rates differ across channels and transaction types?	Can highlight operational patterns that may need attention
5	Which transaction types contribute the most to overall transaction volume and value?	Helps management understand the transaction mix
6	How do customer characteristics and digital engagement relate to transaction activity?	Helps connect customer behaviour with actual transaction activity

7	What transaction patterns are associated with the Risk_Review_Flag?	Supports risk-related monitoring and further investigation
8	How does a customer's preferred banking channel compare with the channel they actually use?	Helps compare stated customer preference with actual behaviour

8. KPI Planning

To support the analytical questions, I proposed eight KPIs that can be calculated from the available data.

KPI	Definition	Why It Matters	Required Data
Transaction Volume	Number of unique transactions	Shows the overall level of transaction activity	Transaction_ID
Total Transaction Value	Sum of transaction amounts	Shows the total monetary value processed	Amount_NGN
Average Transaction Value	Total transaction value divided by transaction volume	Shows the average size of a transaction	Amount_NGN, Transaction_ID
Transaction Success Rate	Successful transactions ÷ total transactions × 100	Helps monitor transaction reliability	Transaction_Status, Transaction_ID
Transaction Failure Rate	Failed transactions ÷ total transactions × 100	Helps identify possible operational issues	Transaction_Status, Transaction_ID

Active Customer Count	Number of distinct customers with at least one transaction in the selected period	Measures customer participation	Customer_ID, Transaction_DateTime
Channel Adoption Share	Transactions through each channel ÷ total transactions × 100	Shows how transaction activity is distributed across channels	Channel, Transaction_ID
Risk-Review Rate	Transactions flagged for review ÷ total transactions × 100	Helps monitor activity requiring further risk review	Risk_Review_Flag, Transaction_ID

These KPIs are intended to give FinTrust a simple overview of transaction activity, customer participation, operational performance, channel usage and risk-review activity.

Before building the final dashboard, the exact calculation rules will need to be kept consistent. For example, I will need to decide how reversed or pending transactions should be treated in some monetary calculations and clearly document that choice rather than make an assumption silently.

9. Dashboard Planning

FinTrust Business Intelligence Dashboard Wireframe

The wireframe below shows the proposed structure of the FinTrust dashboard. It is intended as a planning layout for the Power BI dashboard that will be developed in later weeks.

FINTRUST BUSINESS INTELLIGENCE DASHBOARD					
Filters: Date Customer Segment Channel Transaction Type Transaction Status					
Transaction Volume	Total Transaction Value	Active Customers	Success Rate	Failure Rate	Risk-Review Rate
TRANSACTION VOLUME & VALUE OVER TIME [LINE CHART] Shows how transaction activity and value change over time					
TRANSACTION VALUE BY CUSTOMER SEGMENT			CHANNEL USAGE		
[BAR CHART]			[BAR CHART]		
TRANSACTION STATUS BY CHANNEL			TRANSACTION MIX BY TYPE		

[STACKED BAR CHART]	[BAR CHART]
RISK-REVIEW RATE BY TRANSACTION TYPE [BAR CHART]	PREFERRED CHANNEL VS ACTUAL CHANNEL [MATRIX / BAR CHART]

The dashboard will mainly be designed to answer four things:

What is happening?

Overall transaction volume, value, active customers and processing performance.

How is activity changing?

Transaction trends over time.

What is driving the activity?

Customer segments, channels and transaction types.

Where should FinTrust look more closely?

Transaction failures, reversals and risk-review patterns.

The proposed filters are: Date, Customer Segment, Channel, Transaction Type, and Transaction Status.

10. Initial Week 2–4 Analysis Plan

Week	Main Focus	What I Plan to Do
Week 2	Data preparation and exploratory analysis	Prepare the datasets, handle missing values appropriately, create useful time fields, explore distributions and relationships, and calculate initial KPIs
Week 3	Deeper analysis and dashboard development	Investigate the analytical questions, compare customer segments, channels and transaction types, finalise KPI calculations and build the Power BI dashboard
Week 4	Validation and final reporting	Check calculations, test dashboard filters, refine the visuals, review findings, document limitations and prepare the final presentation/report

In Week 2, I plan to spend more time exploring the patterns behind the questions identified in Week 1 rather than just describing the structure of the data.

Week 3 will be where the dashboard wireframe begins to turn into the actual Power BI dashboard.

Week 4 will focus on checking that everything is accurate, clearly explained and ready to present.

11. Risks and Limitations

There are a few things I will need to keep in mind as the project progresses.

The missing **Device_Type** and **Location** values may affect analysis that depends directly on those fields. High-value transactions may also influence averages, so median values and the overall distribution should be considered during later analysis.

I will also need to avoid treating correlation as causation. For example, if customers with higher digital engagement scores also transact more often, that would show an association, but it would not prove that the engagement score caused the higher transaction activity.

12. Conclusion

Week 1 gave me a clearer understanding of both the FinTrust business problem and the data available for the project.

I reviewed the customer and transaction datasets, checked their structure and quality, confirmed how they relate through Customer_ID, identified the main data-quality issues, and developed the analytical questions and KPIs that will guide the next stage of the project.

The main outcome of this week is not a finished dashboard or final set of business conclusions. It is a clear foundation for the work that comes next.

In Week 2, I will move into data preparation and exploratory analysis, where I can begin testing the questions identified during this initial assessment.

